Learning Better Dynamics Faster With Structured Object Representations: Project for 6.885

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Abstract

Human development of intuitive physics follows a definite progression, from object representations which are online early through the piecemeal acquisition of the dynamical rules governing different domains [14, 1, 5]. This developmental trajectory may not be accidental – a strong starting point of structured object representations may set the stage for the efficient and effective learning of physical dynamics. We propose a model inspired by this possibility. Our model features a perceptual system, a neural derenderer [30], which learns in supervised fashion to extract objects, properties, and relations from visual images. The resulting structured object representations are fed into a highly flexible dynamics learner, based on Interaction Networks [4], which have previously been shown to be capable of learning a diverse set of dynamical laws, including gravitation, elastic collision, and non-rigid dynamics [4, 29, 23]. We train the neural derenderer and interaction networks and compare the interaction network to a purely neural baseline. Future work in this vein will aim to show that the paired neural derenderer and interaction network learn dynamics more effectively and efficiently than an alternative model trained end-to-end.

1 Introduction

Human infants have an early conception of object that structures their expectations about the physical world [6, 25]. Such object representations likely serve to scaffold further learning about physical dynamics, such as learning about support relations [1], gravity and inertia [14], or magnetism [5].

The current work takes inspiration from this developmental trajectory. We implement a model which leverages early developing (separately trained) object representations in perception to support the efficient but flexible learning of physical dynamics. The architecture pairs a neural derenderer [31], trained to deliver representations of objects and attributes, with an interaction network [4] capable of learning a wide variety of natural dynamics.

2 Task

We task our system with detecting violations of expectation (VoE) [24] in an observed video of a physical scene. This requires learning to assign a high likelihood to physically possible scenes and a low likelihood to physically impossible scenes. In keeping with the constraints on human learning we only train our models on possible scenes. This means that discrimination between possible and impossible scenes cannot be used as a training signal. Instead, we train the model to predict likely dynamics from previous observations. We then compare the predictions with posterior observations and, when the difference is high enough, we classify the video as impossible.

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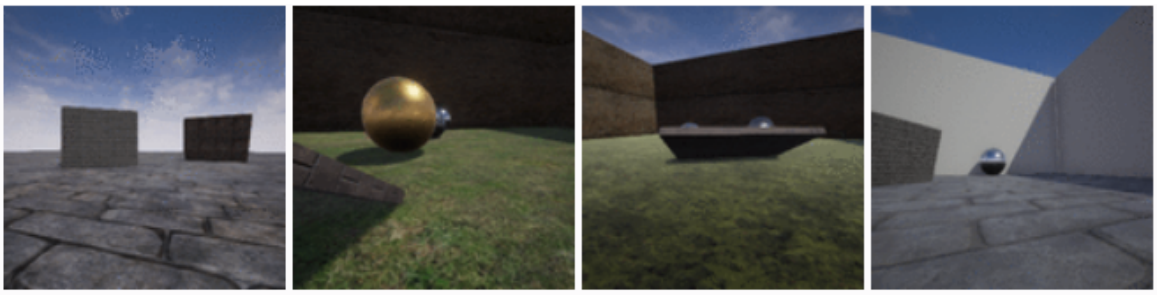

Figure 1: An example scene from IntPhys Benchmark. The task is to deliver a scalar representing the physical plausibility of a given scene (see [20])

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Key Metric since our model predicts a likelihood of the scene being impossible/possible, an Receiver Operating Characteristic metric would be the best quantitative measure.

3 Related Work

3.1 Graph Networks, Interaction Networks, and Visual Interaction Networks

Graph Neural Networks are modular systems for graph-to-graph computation. These networks take in a representation of the nodes, edges, or global structure of a graph and output an updated representation. This general structure has broad applicability. For example, a graph network might be used to segment images into objects (nodes) and relations such as "next to" (edges), or parse atoms (nodes) and bonds (edges) to compute the probability that a molecule is a drug.

In their most general form, GNNs take all input edges into a node i and compute a latent representation for i and its edges. This representation is input into a computation of the activity at a connected node j, and so on recursively for a fixed number of steps or until every node in the graph has been influenced by every other. GNNs have been successfully deployed in many domains for which arbitrary (i.e. non-architecturally fixed) relational inductive biases are useful, including segmentation and classification [28, 13], next frame prediction [7], prediction of traffic patterns [15], and prediction of chemical properties of molecules [8, 10] (see [3] for a review).

GNNs have been fruitfully applied in learning to anticipate the physical dynamics of objects. One example, Interaction Networks [4], learned graph manipulations over object-nodes encoding the dynamics of several kinds of interactions, including gravitational attraction in N-body problems, the elastic collisions of billiard balls, and the non-rigid dynamics of falling strings (represented with nodes at points along a string). Interaction networks were shown to be able to generalize to differing numbers of objects. Visual Interaction Networks (VINs) build on this line of work by appending to Interaction Networks a neural perceptual system [29]. VINs are capable of learning gravitational, elastic, and non-rigid dynamics (similar to the original Interaction Network) from raw pixels.

VINs offer a natural competitor for our model. Like our model, VINs have a powerful dynamics learner based on Interaction Networks at their core. Unlike our model, they are trained end-to-end, from pixels, with many fewer inductive biases. Because they are trained end-to-end, VIN's perceptual systems are unlikely to learn task-general object representations. The lack of powerful, task-independent object representations and the concomitant requirement to use a pixel-based loss present challenges to data efficiency when training on new data sets or learning novel dynamics; handicaps that human learners do not face. These weaknesses might explain why VINs have as of yet been confined to simple (2-Dimensional, high contrast) perceptual domains. Our aspiration is to apply our model to the significantly more realistic (3-Dimensional, w/ non-trivial shapes and textures) domains, such as the data set IntPhys (See Section 5).

3.2 Stochastic Physics Engines

A very different approach to modeling knowledge of physics comes from computational cognitive science. This approach posits a similarity between human psychological representation of physical

scenes and representation in a stochastic physics engine. Physics engines, like those that support engagement with a virtual environment in many contemporary video games, represent space, objects, properties, and forces symbolically and update these representations according to explicit dynamical laws, such as those governing acceleration due to gravity or elastic collisions. When physics engines are programmed with random choices they can be used to define implicit distributions over ways the world might be and ways it might evolve. Random choices allow a stochastic physics engine to encode uncertainty about variables such as object location, velocity, or the direction of a force. These representations and the distributions defined over them can, in turn, be used for probabilistic prediction, inference, and counter-factual evaluation.

The richness of their physical representations, and the computations they support, has made stochastic physics engines a useful model for a diverse set of human intuitive physical reasoning abilities. Physics engines have been invoked to explain infants use of motion cues to object continuation, of shape cues to object identity, and expectations about the behavior of solids and non-solids (see [26] for a review), as well as to quantitatively model infants' relative levels of surprise upon viewing possible or impossible scenes [24] and adults counter-factually informed judgments about stability [2] and causation [9].

One key challenge to the psychological plausibility of stochastic physics engines as models of intuitive physics are the computational requirements that physics engines impose. These computational requirements can be broken down hierarchically into the challenge of doing inference *in* models and that of doing inference *over* models. One notable line of work has attempted to address the first of these challenges using 'neural amortization' or 'inference compilation' methods. These approaches use a symbolic, generative physics engine to train a discriminative, neural model to mimic the input-output patterns that would result from actually computing inference in the generative model. This line of work has attained impressive results, including the rapid recovery of physical scene attributes [30] and quantitative fits to both psychological and neural data [35, 24].

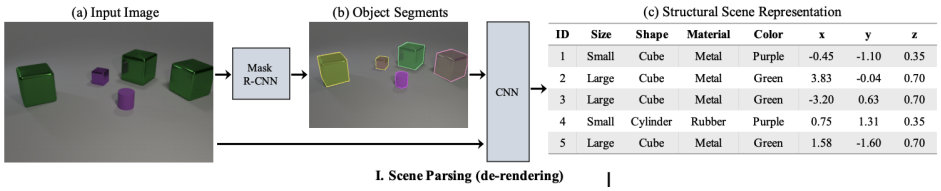
While amortized inference addresses many of the issues related to online processing and inference *in* a model, the challenges of learning physical dynamics, or inference *over* models, remain daunting. Learning in a space of symbolic programs entails search through a vast combinatorial space of programming primitives, often without meaningful gradients to follow. The challenges of such learning are likely orders of magnitude harder than those solved by the neural amortized inference techniques discussed above (The problem that amortized inference resolves, inference *in* a model, is a sub-routine in inference *over* models). Since human developmental trajectories suggest that humans do in fact learn novel physical dynamics, the search is on for a representational medium that will afford such learning (see 3.1 above).

3.3 Inspiration for the Current Approach

In contrast to both Visual Interaction Networks 3.1 and stochastic physics engines 3.2, human cognition seems to strike a balance between data-efficiency, flexible online reasoning, and learning novel physics. One way this compromise might be accomplished is by way of a strong innate starting point supporting perception of objects coupled with a flexible learning system for acquiring dynamics. Since the dynamical rules governing support [1], gravity and inertia [14], magnetism [5], and even magic [17] all operate over objects, a learner whose perceptual system delivers these representations can be expected to learn more quickly across these various domains. In contrast, the process of learning dynamics over these objects should be flexible, relatively unbiased, and capable of learning without gratuitous search. By integrating both interaction networks and amortized object perception, our model aims to strike such a compromise.

4 Model

Our model is mainly composed of a perception module and a dynamics module. The perception module takes each frame of a video and outputs physical attributes such as shape or 3d position of each object at each frame. The perception model is instantaneous, treating each frame independently and outputting an independent graph, possibly with objects in a different order. The dynamics model takes the output of the perception module at each frame and its main output would be the likelihood of the scene being impossible/possible. The plausibility likelihood is computed as the distance between a predicted trajectory and the observed trajectory.



ID	Size	Shape	Material	Color	x	y	z
1	Small	Cube	Metal	Purple	-0.45	-1.10	0.35
2	Large	Cube	Metal	Green	3.83	-0.04	0.70
3	Large	Cube	Metal	Green	-3.20	0.63	0.70
4	Small	Cylinder	Rubber	Purple	0.75	1.31	0.35
5	Large	Cube	Metal	Green	1.58	-1.60	0.70

Figure 2: Figure describing the architecture of the perception module, taken from Neuro-Symbolic-VQA [16]

4.1 Neural perception

The overall functionality of our perception module (Figure 2) consists of taking an image or depth map, as an *input*, x_t, and outputting physical attributes of each object present in the scene, o_t^i: e.g. 3d position, shape, color, etc.

To do so we use a tandem of two submodules:

- Detector: A modern implementation Mask R-CNN [12] [1] that takes the *input*, x_t and outputs *bounding boxes* for each *object* in the scene, b_t^i
- De-Renderer: A customized version of a ResNet-34 [11] that takes the *input*, x_t and the predicted bounding box of an *object*, b_t^i and outputs the physical attributes of such *object*, o_t^i. [2]

4.2 Dynamics Network

The dynamics component of out model is realized with an Interaction Network [4] (see 3.1 above) which serves as a learnable physics engine. The model takes graphs as input, performs object- and relation-centric reasoning in a way that is analogous to a simulation, and is implemented using deep neural networks. Mathematically, the input can be represented as a graph $G\langle O, R\rangle$, where the nodes, O, correspond to the objects, and the edges, R, to the relations. For a system with N_O objects and N_R relations, the inputs to the Interaction Network are:

$$O = \{o_j\}_{j=1\cdots N_O}, \quad R = \{\langle i, j, r_k\rangle_k\}_{k=1\cdots N_R}, \text{ where} i \neq j, 1 \leq i, j \leq N_O, \quad X = \{x_j\}_{j=1\cdots N_O}$$

The O represents the states of each object.The triplet $\langle i, j, r_k\rangle_k$ represents the k-th relation in the system, from sender o_i, to receiver o_j, with relation attribute r_k. The X represents external effects, such as active control inputs or gravitational acceleration, which are defined as not being part of the system, and which are applied to each object separately.
The interaction network can be expressed as:

$$IN(G) = \phi_O(a(G, X, \phi_R(m(G))))$$

$$\begin{aligned} m(G) &= B = \{b_k\}_{k=1\cdots N_R} & a(G, X, E) &= C = \{c_j\}_{j=1\cdots N_O} \\ f_R(b_k) &= e_k & f_O(c_j) &= p_j \\ \phi_R(B) &= E = \{e_k\}_{k=1\cdots N_R} & \phi_O(C) &= P = \{p_j\}_{j=1\cdots N_O} \end{aligned}$$

The function m rearranges the object and relations into interaction terms, $b_k = \langle o_i, o_j, r_k\rangle \in B$, one per relation, which correspond to each interaction's receiver, sender and relation attributes. The relational model, ϕ_R, predicts the effect of each interaction, $e_k \in E$, by applying f_R to each b_k. The aggregation function, a, collects all effects, $e_k \in E$, that apply to each receiver object, merges them , and combines them with O and X to form a set of model inputs, $c_j \in C$, one per object, The object model ϕ_O, predicts how the interactions and dynamics influence the objects by applying f_O to each c_j, and returning the results, $p_j \in P$. This basic IN can predict the evolution for states in a dynamical system, for physical simulation, P may equal the future states if the objects, O_{t+1}.

[1] we used Detectron2, [32]

[2] We use the bounding box to filter the relevant part of the image, then we concatenate the original image and use this as the input to the De-Renderer.

4.3 Bi-directional RNN

We use a Bi-directional RNN, in particular a multilayered Bi-directional GRU, as a learnable dynamics engine. Intuitively, the RNN's hidden state models the current state of all objects in a scene, including occluded ones, whereas the input receives only the (possibly noisy) observed physical attributes at each frame. We specifically use a Bi-directional model because the current task requires backward object tracking: e.g. if an object gets revealed behind an occluder, the model should be able to reason that such object position should have been behind the occluder before observing it. The combination of bidirectionality and memory pertain fundamentally to intuitive physics. They together smooth perception noise across frames and enable the model to predict object states even under occlusion.

Training Method Only on possible scenes: take the output of the perception module for all frames in a scene $O = \{Obs_t^o\}_{t=1,o=1}^{T,N_{max}}$ and train the network to predict the ground truth trajectory of all objects:

$$\left\{\left(s_{t,o}, \boldsymbol{\psi}_{t,o}, \mathbf{x}_{t,o}, \boldsymbol{\omega}_{t,o}, v_{t,o}, e_{t,o}\right)\right\}_{t=1,o=1}^{t=T,o=N_{max}}$$

Note that besides the symbols mentioned in a previous section we also train the network to predict the visibility, $v_{t,o}$, and existence, $e_{t,o}$, of the objects. Visibility, $v_{t,o}$, will help the network reason about potential data association issues, only using observations to update visible objects. Whereas existence, $e_{t,o}$, and having a fixed number of objects N_{max}, will help to fix the input/output dimensionalities of the RNN.

Inference Method At inference time, we input the sequence of observations given by the perception module and then use the trained RNN to get the state of each object at all frames (even when not visible). Since the network was trained only possible scenes, the sequence of observations belonging to possible scenes would be in-sample, therefore yielding a consistent ground truth trajectory for each object. On the other hand, on impossible scenes, the observations include some permanent inconsistency, such as an object disappearing behind an occluder. In this case out-of-sample observations would yield inconsistent output trajectories, to a point that it should be easy to heuristically classify it as impossible.

Possible Failing Reasons RNNs are designed with a fixed-dimensional input and hidden state vectors, whereas in this case we have a dynamic number of objects in both hidden and observation spaces. Moreover, RNNs are not permutation invariant, meaning that the order of the input features changes the output in an unpredictable way, but in an object-centric representation, the order of the objects should not have a significant effect in the predicted state of such objects. In general this creates a data association problem between the hidden state and the observations.

Fundamentally solving these problems would be the scope of future work, however, for the sake of this paper one could try some heuristics:

- i) have a fixed number of objects in both hidden and observation spaces and create an "existing" flag that tells the model which objects actually exist. This makes it possible to flatten out these matrices (number of objects $\times$ number of attributes) and get a fixed dimensional vector.
- ii) lexicographically organize the objects in both output and input of the RNN to ease out the data association.

however, these heuristics still have flaws, and could pose a high risk of preventing the network from learning anything at all.

5 Data Sets

We use the intphys dataset [20] which is commonly used as a a benchmark for visual intuitive physics reasoning. The intphys benchmark can be applied to any vision system, engineered, or trained, provided it can output a scalar when presented with a video clip, which should correspond to how physically plausible the video clip is. The test set contains well-matched videos of possible versus impossible events, and the metric consists of measuring how well the vision system can tell apart the possible from the impossible events. The dataset is divided into four sets:

1. **Train dataset** includes 15k scenes with full metadata. Importantly, all videos are possible.
2. **Test dataset** Test datasets include 1080 scenes for each of four blocks without metadata. Each block measures one of the following principles of intuitive physics: block O1 - object permanence, block O2 - shape constancy, block O3 - spatio-temporal continuity and block O4 - energy conservation.
3. **Development dataset** includes 30 videos for each block with partial metadata (physical plausibility of the scene is given).
4. **Dev-meta dataset** contains 72 videos with full metadata. It includes both possible and impossible videos which are testing principles from one of the first three blocks.

6 Experiments

6.1 Dynamics prediction with an Interaction Network

Model specification We used the train split of the intphys dataset to train an interaction network. We kept aside 30% of this data for validation. We used the ground truth physics symbols, which are available in the train split, to featurize the objects. In particular, an object is characterized by a vector of 25 values: 2 are used for one-hot encoding whether an object is present or not (if not, it is a dummy object), 2 are used to one-hot encode whether an object is an occluder or not, 2 are used to indicate if an object is visible or not, 3 for the location (x,y,z), 3 for the velocity (x,y,z), 3 for the scale (x,y,z), 3 for the rotation (roll, pitch, yaw), 1 for speed (occluder only), 1 for friction, 1 for mass and 4 are used to one-hot encode the shape. The input to the interaction network is information about one video frame, which includes four matrices: objects, sender relations, receiver relations, relation info. The objects matrix has dimensions 5×25 where 5 is the maximum number of objects in any video and 25 is the object feature vector. The sender and receiver relations are binary matrices of size 5×20 (20 because each of the 5 objects can interact with the remaining 4 i.e. it is the number of edges in a fully connected graph) which encode that every object interacts with every other object, i.e. every object is both a sender and a receiver for all the other objects. The relation info is not relevant for our task so we fill it with all zeros. The effect vector size is set to 100. The target is the objects matrix for the next timestep, which encodes objects' state after one step.

Training method We trained the model for 200 epochs and kept the checkpoint with the best validation results. We used Adam optimizer with a learning rate of 0.001. The loss is an aggregation of a Cross-Entropy Loss for the categorical variables and a Mean Squared Error for the continuous variables. The losses from non-existent objects (dummy objects) did not contribute to the loss.

6.2 Results

We used our trained interaction network to perform one-step prediction in the future, such as shown in figure 3. We see that the predicted location and rotation of the cube matches well the ground truth. Importantly, we note that this is in the case where we use the ground truth state at timestep t to predict the state at timestep $t + 1$. If we use the prediction at time t to predict $t + 1$, the predictions suffer from an accumulation of errors which can grow exponentially large. Indeed, if we plot the Mean Absolute Error (MAE) of the predictions and ground truth locations at each timestep t, we notice that the exponential trend of the error in figure 6.2, which shows up as a linear line in log scale. On the other hand, the error is generally really low when we use a one-step prediction, as seen in Figure 6.2. Note that the locations can have absolute values in the order of hundreds, so an absolute error of 0-20 is pretty low.

We also evaluated the predictive power of our trained network on the dev-meta set of the intphys dataset by detecting anomalous scenes in the videos. To do this, we again used one-step prediction and to ensure object permanence and shape constancy we set the predicted existence bit and the predicted shape to be the same as in the previous step. A video is considered anomalous if the shape of the object has changed or any of the x, y, z locations are off by more than 200 units. With these specifications we obtained perfect accuracy on the dev-meta. While this might look surprising at

first, it is important to remember that using the ground truth symbols makes the predictive task much easier than it would otherwise be. For example, we do not need to worry about objects that are not visible (due to a position out of view or behind an occluder), because we would still have the ground truth locations. We also do not need to perceptually detect the shape and other attributes which are crucial for the prediction task. This means that while the model shows promising results, the ability to apply it in real-world scenes would require a powerful perceptual network and infrastructural setup to handle the special cases, such as those when an object is not visible.

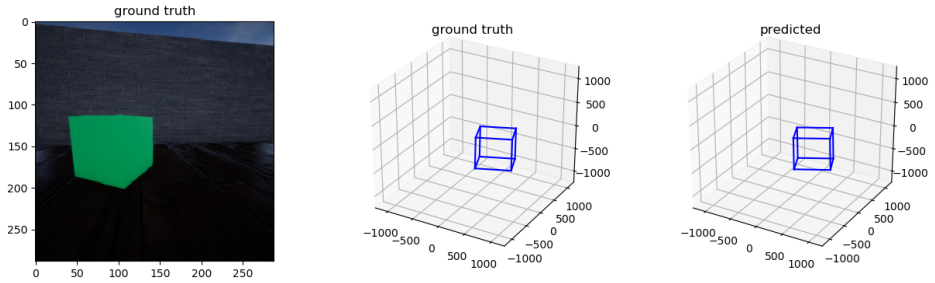
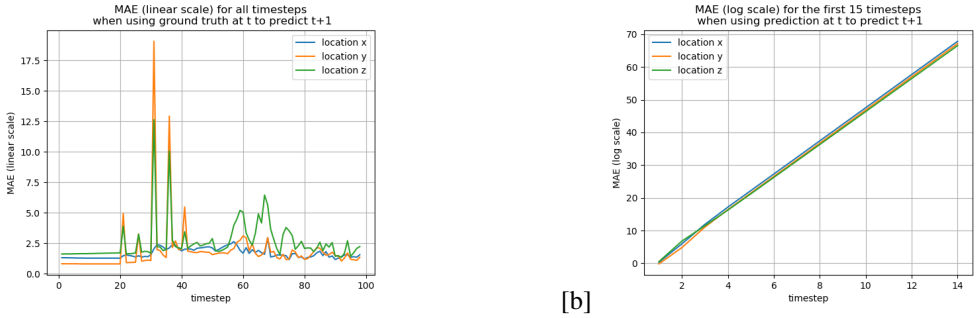


Figure 3: Example ground truth object state at frame 13 (the first two plots) and predicted object state when using ground truth state at time t to predict $t+1$.



[a] [b]

Figure 4: a) MAE (linear scale) for all timesteps when using ground truth at t to predict t+1 and b) MAE (log scale) for the first 15 timesteps when using prediction at t to predict t+1

7 Contributions

Our model offers a novel approach to the problem of learning intuitive physics which mirrors the developmental trajectory seen in human infants. Our model also features an application of Interaction Networks to a more realistic data set, featuring 3-Dimensional objects and complex attributes such as shape, then had been previously demonstrated. Our Dynamics Network successfully learned to simulate a physics step given the ground truth objects attributes from the previous step when trained on data from the IntPhys data set.

The current implementation fell short of our aspirations in several ways however.

(1) Our perceptual module is not occluder-resistant. That is, since the perceptual module takes in an input scene and outputs a graph describing it, objects which have gone behind occluders are essentially non-existent for this module.

(2) Our dynamics network is trained on ground truth, so any disappearance or appearance of an object at test time will be treated as an impossibility (i.e. if it is getting its messages from the perceptual system)

(3) This could be addressed if the dynamics network could be trained to predict the next timestep $t+1$ from both an input observation *and* the model's conclusions about the underlying hidden state of the world at the time t, essentially reproducing the same trick as an RNN. Unfortunately the Interaction Networks framework does not make this straightforward.

(4) Finally, Interaction Networks face a data association problem. That is, although the network is invariant to the order of the objects, our task would require recognizing that the n-th object at timestep t corresponds to the n-th object at timestep $t+1$ in order to detect anomalies (inconsistencies in the timeline) using representations of instantaneous scene graphs. This problem is exacerbated by the perceptual system's difficulty with occlusion, as the perceptual system presents the dynamics network with an unequal number of objects at different timesteps, due to objects becoming visible or invisible.

8 Limitations and Future Work

Apart from the more immediate challenges to implementation discussed in 7, our architecture faces several theoretical limitations. Many of these weaknesses can be seen as the flip-side of the model's strengths. Our perceptual system, for example, delivers representations of objects and their attributes whose structure could promote efficient learning of dynamics. In order to deliver these representations however, our perceptual derenderer requires ground-truth supervision for objects and attributes. Real world learners are unlikely to have such a luxury. One long-term goal might be to find a more realistic compromise between innate knowledge and learning (perhaps coupling dynamical learning with innate symbolic knowledge of objects in the form of a graphics engine, but not a physics engine).

Similarly, our dynamics network is a flexible learner of novel phyiscal dynamics. But this flexibility also engenders weaknesses in data efficiency and the task-specificity of learned representations. To take just one example, one strength of *stochastic physics engines* that has been key to their use in models of human intuitive physics is their ability to support counterfactual reasoning [2]. It is unclear whether our dynamics network, which is trained on observational data alone and learns only implicit representations of dynamics, can support such reasoning. It is possible that the graph representations over which our dynamics network operates will support certain kinds of counterfactuals ('what if the forces on this object had been doubled?'), but not others ('what if the force of gravity decayed as function of distance cubed?' or 'What if there were gravitational attractors *and also gravitational repulsors?*'). Future work might examine Interaction Networks' ability to learn and handle such counterfactuals, perhaps by training them in an environment that permits interventions.

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